

IGCSE 0478 Pseudocode Skills — Answer Key

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Full coding-format answers and the complete MCQ answer key for every quiz on the practice site.

Companion file to [index.html](#).

Part 1 — Coding Format Answers

▼ Basics Answers

```
DECLARE Amount, TotalPrice, Discount, FinalPrice : REAL
```

```
Amount <- 100
```

```
TotalPrice <- Amount * 3.5
```

```
Discount <- 0.2
```

```
FinalPrice <- TotalPrice - TotalPrice * Discount
```

```
OUTPUT "Final price is ", FinalPrice
```

```
TriangleArea <- Base * Height / 2
```

```
OUTPUT "Triangle area is ", TriangleArea
```

► Selection Answers

► Loops Answers

► Standard Methods Answers (Totalling, Max, Min, Linear Search, Bubble Sort)

► Validation / Testing Answers

► Functions and Procedures Answers

Part 2 — MCQ Answer Key

1. Basics

Which keyword declares a variable?

- INPUT
- ✓ **DECLARE**
- OUTPUT
- NEXT

DECLARE states variables and data types.

Best data type for a name?

- INTEGER
- REAL
- ✓ **STRING**
- BOOLEAN

Names are text, so use **STRING**.

What does <- mean?

- Equal to
- **✓ Assignment**
- Not equal
- End

<- assigns a value.

Best data type for whole-number age?

- **✓ INTEGER**
- REAL
- STRING
- CHAR

Whole numbers use INTEGER.

Why does sequence matter?

- It changes names
- It removes declarations
- It avoids loops
- **✓ Order affects result**

Algorithms run in order.

2. Selection

Which keyword ends IF?

- NEXT
- ENDWHILE
- **✓ ENDIF**
- STOP

ENDIF closes IF.

Both conditions true?

- OR
- **✓ AND**
- NOT
- OTHERWISE

AND requires both.

Best for menu choices?

- **✓ CASE OF**
- DECLARE
- OUTPUT
- LENGTH

CASE OF handles fixed choices.

Not equal operator?

- =
- <=
- >=
- **✓ <>**

<> means not equal.

Invalid CASE choice?

- NEXT
- THEN
- **✓ OTHERWISE**
- INPUT

OTHERWISE handles other values.

3. Loops

Exactly 10 repetitions?

- **✓ FOR**
- CASE
- IF
- DECLARE

FOR is count-controlled.

Always runs once?

- FOR
- WHILE
- **✓ REPEAT UNTIL**
- CASE

REPEAT UNTIL tests at the end.

May run zero times?

- REPEAT
- ✓ **WHILE**
- OUTPUT
- CASE

WHILE tests at the start.

Closes FOR loop?

- ENDIF
- ENDCASE
- UNTIL
- ✓ **NEXT Counter**

NEXT closes FOR.

Sentinel value?

- ✓ **Stop value**
- Data type
- Declaration
- Trace column

A sentinel stops input.

4. Standard Methods

Total starts at?

- **✓ 0**
- 1
- 100
- TRUE

Totals usually start at 0.

Increment counter?

- $\text{Count} \leftarrow \text{Count} - 1$
- **✓ $\text{Count} \leftarrow \text{Count} + 1$**
- $\text{Count} \leftarrow 0$
- $\text{Count} \leftarrow \text{Total}$

Add 1 to Count.

Average formula?

- $\text{Total} * \text{Count}$
- $\text{Count} / \text{Total}$
- $\text{Total} + \text{Count}$
- **✓ $\text{Total} / \text{Count}$**

Average is total divided by count.

In a linear search, the flag Found starts as?

- TRUE
- **✓ FALSE**
- 1
- 100

Found starts FALSE and only becomes TRUE when the item is located.

To find the highest value in a list, Highest should be set to...?

- 0
- **✓ The first value in the list**
- 1000
- The last value in the list

Starting at the first value works for every list, including one made only of negative numbers.

Which comparison is used when finding the LOWEST value?

- IF Number > Lowest
- IF Number = Lowest
- **✓ IF Number < Lowest**
- IF Number <> Lowest

Finding the minimum uses < ; finding the maximum uses > .

4a. Linear Search (dedicated quiz)

What causes the search loop to stop early, before reaching the end of the list?

- The counter reaches zero
- **✓ The Found flag becomes TRUE**
- The array runs out of memory
- The list is sorted first

Found is set to TRUE as soon as a match is located, which ends the loop immediately.

What data type must Found be declared as?

- INTEGER
- STRING
- **✓ BOOLEAN**
- CHAR

Found can only hold TRUE or FALSE, so it is a BOOLEAN.

In the worst case, how many comparisons does a linear search need for a 40-item list?

- 1 comparison
- 20 comparisons
- 0 comparisons
- **✓ 40 comparisons**

If the item is last or missing, every one of the 40 items must be checked.

4b. Bubble Sort (dedicated quiz)

What happens to Last after each complete pass through the list?

- **✓ It decreases by one**
- It increases by one
- It resets back to one
- It stays exactly the same

Each pass places the next-largest value correctly, so Last is reduced by one to skip it next time.

What condition ends the REPEAT...UNTIL loop in the sample bubble sort?

- Index reaches First
- The array becomes empty
- **✓ No swap was made or Last equals one**
- Temperature exceeds one hundred

The loop stops once a full pass makes no swaps, or Last has been reduced to one.

Why are two neighbouring elements swapped during a pass?

- Because they are exactly equal
- **✓ Because they are out of the correct order**
- Because the loop has just finished
- Because Last has reached zero

A swap happens whenever an element is greater than the one next to it, fixing the order.

5. Validation and Verification

Value between limits?

- Presence
- **✓ Range**
- Length
- Double entry

Range check tests lower and upper limits.

Blank input check?

- ✓ **Presence**
- Range
- Length
- Bubble

Presence check prevents blanks.

Exactly 8 chars?

- Range
- ✓ **Length**
- Double entry
- Trace

Length check counts characters.

Data entered twice?

- Range
- Length
- ✓ **Double entry**
- CASE

Double entry verifies matching inputs.

Valid mark 0-100 rejects?

- 50
- 0
- ✓ **101**
- 100

101 is outside the range.

6. Testing and Debugging

Normal data means?

- ✓ **Expected valid data**
- Invalid data
- Boundary outside
- No data

Normal data should be accepted.

Extreme values for 0-100?

- -1 and 101
- ✓ **0 and 100**
- 50 and 60
- 0 and 101

Extreme values are valid ends.

Boundary for lower end 0?

- ✓ **-1 and 0**
- 0 and 1
- 99 and 100
- 100 and 101

Boundary checks just outside and just inside.

Trace table records?

- ✓ **Variable changes**
- Only headings
- Only code
- Only comments

Trace tables record changing values.

Missing keyword after IF?

- NEXT
- ✓ **ENDIF**
- ENDCASE
- DECLARE

IF blocks end with ENDF.

7. Functions and Procedures

What is the key difference between a FUNCTION and a PROCEDURE?

- ✓ **A FUNCTION always returns a value; a PROCEDURE does not**
- A PROCEDURE always returns a value; a FUNCTION does not
- They are exactly the same
- A FUNCTION cannot take parameters

A FUNCTION always RETURNS a value; a PROCEDURE performs a task without returning one.

How do you invoke a PROCEDURE named PrintReport?

- PrintReport()
- ✓ **CALL PrintReport()**
- RETURN PrintReport()
- OUTPUT PrintReport()

PROCEDURES are invoked with the CALL command.

Which keyword must appear in a FUNCTION header but never in a PROCEDURE header?

- CALL
- ✓ **RETURNS**
- DECLARE
- ENDFUNCTION

RETURNS specifies the data type of the value a FUNCTION sends back.

Which statement sends a value back to the calling code inside a FUNCTION?

- OUTPUT
- CALL
- ✓ **RETURN**
- INPUT

RETURN passes the result back and ends the FUNCTION.

A parameter declared BYREF instead of BYVAL means...?

- The parameter is optional
- ✓ **Changes made inside the subroutine also change the original argument**
- The parameter must be a STRING
- The subroutine cannot use the parameter

BYREF passes a reference so changes affect the original variable; BYVAL passes a copy.

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